

Jenkins

Class no:1

Jenkins

Jenkins is an open-source automation server that helps developers build, test, and deploy their software.

Think of it as the ultimate project coordinator for software development. Instead of a human having to manually compile code, run tests, and push updates to a server every time a change is made, Jenkins automates the entire sequence.

Here is a breakdown of how it works, why it's so popular, and where it fits in the development lifecycle.

The Jenkins will be installed default in **/Jenkins/var/lib/Jenkins**

Default port number: **port 8080**

The Core Concept: CI/CD

Jenkins is the industry standard tool for implementing **CI/CD**, which stands for:

- **Continuous Integration (CI):** Every time a developer commits new code to a repository (like GitHub or GitLab), Jenkins automatically triggers a "build." It compiles the code and runs automated tests to ensure the new changes didn't break anything.
- **Continuous Delivery/Deployment (CD):** If the build and tests pass, Jenkins can automatically deploy the updated application to a staging or production server.

How Jenkins Works (The Pipeline)

Jenkins uses a feature called **Pipelines**, which allows you to define your entire build process as code (usually written in a file called a Jenkinsfile).

A typical Jenkins pipeline follows these steps:

[Code Commit] —> [1. Fetch Code] —> [2. Build/Compile] —> [3. Run Tests] —> [4. Deploy]

1. **Trigger:** A developer pushes code to GitHub.
2. **Build:** Jenkins detects the push, grabs the code, and compiles it (e.g., turning Java code into a JAR file).
3. **Test:** Jenkins runs unit tests and integration tests. If a test fails, it alerts the team immediately.
4. **Deliver/Deploy:** If all tests pass, Jenkins packages the application (perhaps into a Docker container) and pushes it to your servers.

Hands on : Launching and set up of Jenkins Via AWS EC2

Step no:1 Launch an EC2 instance (ubuntu) with a larger size to run Jenkins.

Choose default SG and keep All TCP enabled in it. (default port no. for Jenkins:8080)



Instances (1/2) Info		Last updated 3 minutes ago	Connect	Instance
Saved filter sets Choose filter set ▼				
Find Instance by attribute or tag (case-sensitive)				
Name	Instance ID	Instance state	Instance type	
jenkins-instance	i-0633e617639a60e45	Running	m7i-flex.large	

Step no:2 Jenkins requires java to be installed as prerequisites.

Go to Jenkins installation → Linux → Ubuntu

Installation of Java

Jenkins requires Java to run, yet not all Linux distributions include Java by default. Additionally, not all Java versions are compatible with Jenkins.

The Jenkins will be installed default in **/Jenkins/var/lib/Jenkins**

Default port number: **port 8080**

Copy and paste this code in the Aws console:

sudo apt update

sudo apt install fontconfig openjdk-21-jre

java -version

```
root@ip-172-31-14-119:/home/ubuntu# sudo apt update
sudo apt install fontconfig openjdk-21-jre
java -version
```

Installation of Jenkins

Copy and paste this code in the console after installing the java package.

Long Term Support release

A [LTS \(Long-Term Support\) release](#) is chosen every 12 weeks from the stream of regular releases as the stable release for that time period. It can be installed from the [debian-stable apt repository](#).

```
sudo wget -O /etc/apt/keyrings/jenkins-keyring.asc \
https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
echo "deb [signed-by=/etc/apt/keyrings/jenkins-keyring.asc]" \
https://pkg.jenkins.io/debian-stable binary/ | sudo tee \
/etc/apt/sources.list.d/jenkins.list > /dev/null
sudo apt update
sudo apt install jenkins
```

```
root@ip-172-31-14-119:/home/ubuntu# sudo wget -O /etc/apt/keyrings/jenkins-keyring.asc \
https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
echo "deb [signed-by=/etc/apt/keyrings/jenkins-keyring.asc]" \
https://pkg.jenkins.io/debian-stable binary/ | sudo tee \
/etc/apt/sources.list.d/jenkins.list > /dev/null
sudo apt update
sudo apt install jenkins
```

Start the Jenkins server and check the status is running.

```
root@ip-172-31-14-119:/home/ubuntu# systemctl start jenkins
root@ip-172-31-14-119:/home/ubuntu# systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service;
   Active: active (running) since Mon 2026-06-01 15:26:50 U
   Invocation: bce4554e4981420cb45ab7774a750cb8
```

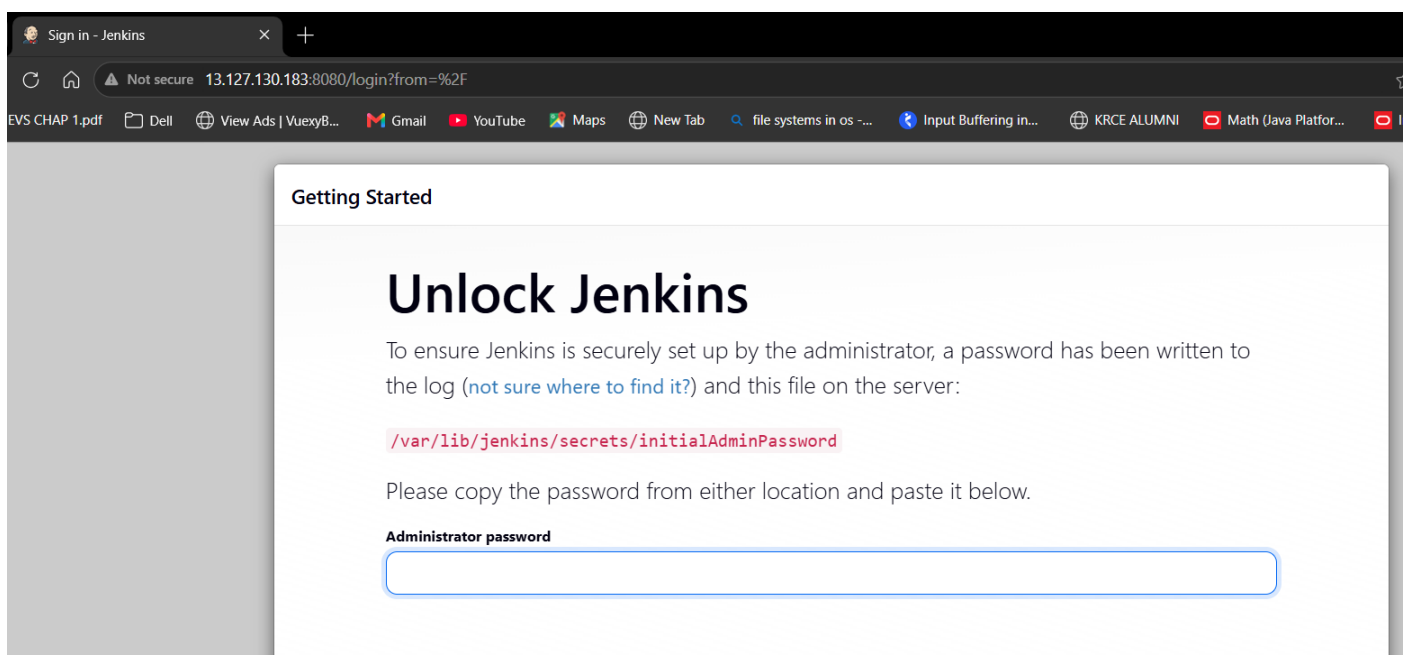
Step no:3 Copy the public IP of the instance:8080 (default port of Jenkins) and paste it in a private browser.

```
No VM guests are running outdated hypervisor
root@ip-172-31-14-119:/home/ubuntu#
```

i-0633e617639a60e45 (jenkins-instance)

PublicIPs: 13.127.130.183 PrivateIPs: 172.31.14.119

It will provide login page for Jenkins and the path where the password is located for login. Copy the path.



Sign in - Jenkins

Not secure 13.127.130.183:8080/login?from=%2F

Getting Started

Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log ([not sure where to find it?](#)) and this file on the server:

```
/var/lib/jenkins/secrets/initialAdminPassword
```

Please copy the password from either location and paste it below.

Administrator password

Step no:4 Paste in the console with Cat command to see the password.

```
root@ip-172-31-14-119:/home/ubuntu# cat /var/lib/jenkins/secrets/initialAdminPassword
488b8a76378c449c8a98c5f1cf653ab0
root@ip-172-31-14-119:/home/ubuntu#
```

Now copy the password and paste it in the Jenkins login page password dialog box and login.

Getting Started

Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, the log ([not sure where to find it?](#)) and this file on the

`/var/lib/jenkins/secrets/initialAdminPassword`

Please copy the password from either location and

Administrator password

Step no:5 It will ask you to install the suggested plugins so that our app works fine without any problem. Install them.

Getting Started

Customize Jenkins

Plugins extend Jenkins with additional features to support many different needs.

Install suggested plugins

Install plugins the Jenkins community finds most useful.

Select plugins to install

Select and install plugins most suitable for your needs.

Getting Started

☒ Folders	☒ OWASP Markup Formatter	☒ Build Timeout	☒ Credentials Binding	☐ Plugin Utilities API
☒ Timestampers	☒ Workspace Cleanup	☒ Ant	☐ Gradle	☐ Font Awesome API
☐ Pipeline	☐ GitHub Branch Source	☐ Pipeline: GitHub Groovy Libraries	☐ Pipeline Graph View	☐ Bootstrap 5 API
☐ Git	☐ SSH Build Agents	☐ Matrix Authorization Strategy	☐ LDAP	☐ Jackson Annotations 2 API
☐ Email Extension	☐ Mailer	☐ Dark Theme		☐ SnakeYAML Engine API
				☐ Woodstox Core API
				☐ Jackson 3 API
				☐ JQuery3 API
				☐ ECharts API
				☐ Prism API
				☐ Display URL API
				☐ Checks API
				☐ JUnit
				☐ Matrix Project
				☐ Resource Disposer
				☐ Workspace Cleanup
				☐ Ant
				☐ Gantt
				☐ JavaBeans Activation Framework

Step no:6 Once the installation is done. The login page ask us to create a username password and our email address. Provide them save and continue.

Getting Started

Create First Admin User

Username

nithya_vel

Password

Confirm password

Full name

Nithya_Sri

E-mail address

nithyavel04@gmail.com

Jenkins 2.555.2

[Skip and continue as admin](#) [Save and Continue](#)

We can see the Jenkins url → click continue.

Getting Started

Instance Configuration

Jenkins URL:

http://13.127.130.183:8080/

The Jenkins URL is used to provide the root URL for absolute links to various Jenkins resources. That means this value is required for proper operation of many Jenkins features including email notifications, PR status updates, and the BUILD_URL environment variable provided to build steps.

The proposed default value shown is **not saved yet** and is generated from the current request, if possible. The best practice is to set this value to the URL that users are expected to use. This will avoid confusion when sharing or viewing links.

Step no:7 We can see the Jenkins home page.

Click on new item

 **Jenkins**

[+ New Item](#)

[📁 Build History](#)

Build Queue

▼

No builds in the queue.

Build Executor Status

0/2 ▼

Welcome to Jenkins!

This page is where your Jenkins jobs will be displ builds or start building a software project.

Start building your software project

Create a job

Set up a distributed build

Set up an agent

Configure a cloud

Learn more about distributed builds

Name the pipeline as : **mypipeline** → type as: **Pipeline** → ok.



Jenkins

/ All



/ New Item

New Item

Enter an item name

mypipeline

Select an item type



Pipeline

Build, test, and deploy using

To visualise how the pipeline helps to build, test and deploy → scroll down to pipeline script.

We can write our own script or pull it from GitHub repo.

I am using a sample pipeline script provide by the Jenkins itself.

Definition

Pipeline script

Script ?

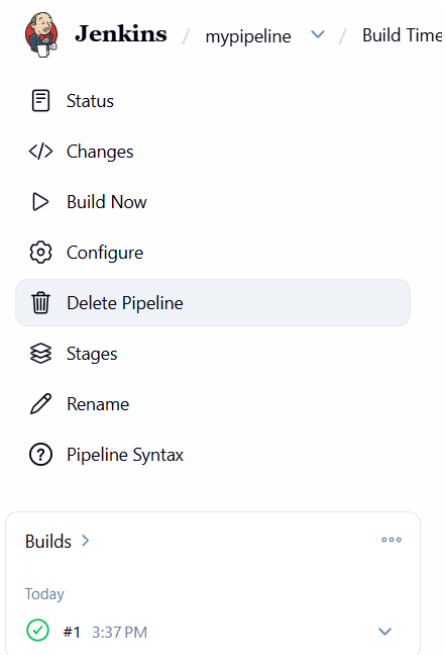
```
1 pipeline {  
2     agent any  
3  
4     stages {  
5         stage('Hello') {  
6             steps {  
7                 echo 'Hello World'  
8             }  
9         }  
10    }  
11 }  
12
```

Save

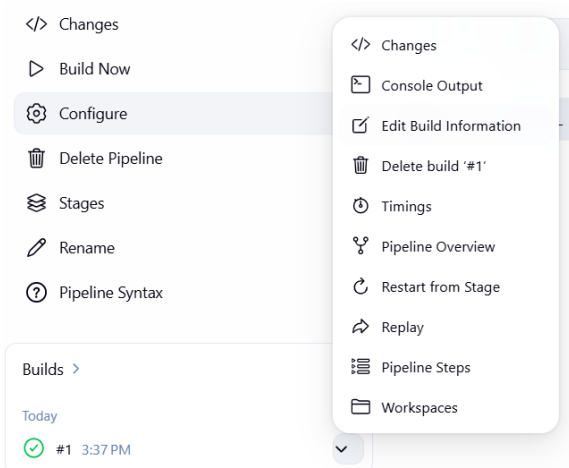
Apply

Choose save.

Step no:8 Now click Build now option. A build will be executed.



Once executed, we need to see the output of Jenkins script through **console output**.



We can see that our script has been executed successfully there.

Console

```
Started by user Nithya_Sri
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/lib/jenkins/workspace/mypipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Hello)
[Pipeline] echo
Hello World
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```